

Chapter 4

Points, Part 2 of 2

In this chapter we mostly develop a better constructive feel for how inversion of a plane with respect to a circle with both positive and negative power works by solving the problems that are converse to the problems that we solved in Chapter 3.

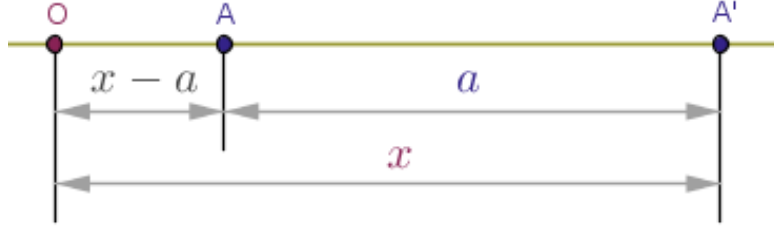
That is. In the upcoming challenges, given two points that are the inverses of each other, we will be asked to recover either the location of the center or the length of the radius of the respective inversion. As usual, all the relevant constructions are to be carried out with the Euclidean tools, a compass and a straightedge, alone.

Problem 6. Given two distinct points A and A' that are the inverses of each other with respect to a circle with a given radius r , locate the center of the corresponding inversion, O , with positive power.

Solution. One way to solve this problem is to investigate the algebraic side of the problem and then translate the found solution that is cast in the language of algebra into a solution that is cast in the language of geometry.

To that end, let the point O be the center of the inversion sought-after.

Then, we are given the distance $|AA'| = a$ between the points A and A' and, as one possible approach, we want to recover the distance, x , between the sought-after center of inversion, O , and one of the given points that is located on the other side of the remaining given point (Figure 4.1):



For example. If the mystery center of inversion, O , is closer to the given point A than it is to the given point A' then x will be the distance between that center of inversion and the point A' as shown in the diagram above: $x = |OA'|$.

Under that sample assumption, according to **Condition 3** of **Definition 1** and our definition of inversion with respect to a circle we positive power, which implies that $|OA| = (x - a)$, we have:

$$x \cdot (x - a) = r^2$$

from where it follows that:

$$x^2 - ax - r^2 = 0$$

Solving the above equation quadratic in x for x , we find that:

$$x_{1,2} = \frac{a}{2} \pm \sqrt{\left(\frac{a}{2}\right)^2 + r^2} \quad (1)$$

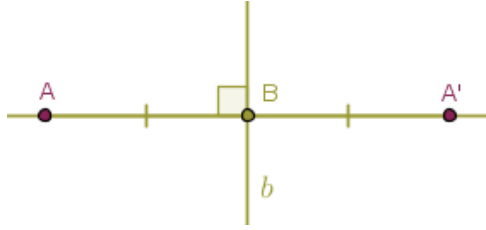
Analyzing the above relation, we see that all the algebraic material in it is constructible with Euclidean tools alone. That is, a geometric equivalent of the square root in (1) is the hypotenuse of a right triangle the lengths of whose sides are given because the length of the radius r is given and, because the length a is given, a geometric equivalent of $(a/2)$ is one half of a line segment of length a and any line segment can be bisected with Euclidean tools alone.

As such, translating the above algebraic or arithmetic sentence into a solution that is cast in the language of geometric objects, we have the following.

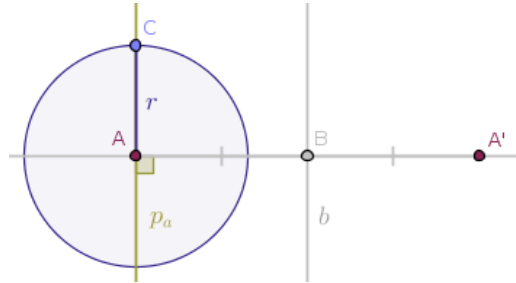
The mystery distance x can be represented, if we take the plus sign in (1), as a sum of one half of the distance a between the given points A and A' and the length of the hypotenuse of a right triangle whose one side is precisely that half of the given line segment a and whose other side is equal in length to the given length of the radius of inversion, r .

Hence, we come up with the following algorithm for locating the center of inversion under the given conditions.

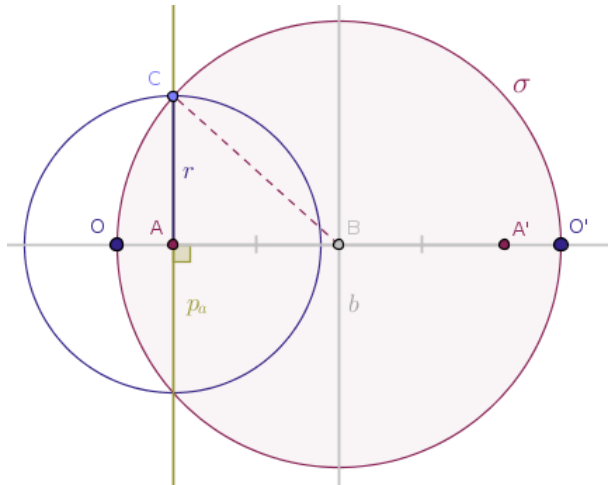
1. Draw a straight line through the given points A and A' , bisect the line segment AA' with a perpendicular bisector, b , in order to locate the point B on the line segment AA' such that $|AB| = |BA'| = a/2$ (Figure 4.2):



2. Draw a straight line, p_a , perpendicular to the line segment AA' through the point A (for example), draw a circle that has the point A as its center and the given distance r as its radius. Let C be one of the points where the straight line p_a and that circle intersect (Figure 4.3):



3. Draw the circle, σ , centered at the point B with the radius $|BC|$ until σ intersects the straight line AA' extended at the points O and O' (Figure 4.4):



The points O and O' thus constructed are the sought-after centers of the two circles of radius r under the inversion with respect to which with positive power, the given points, A and A' , are the images of each other.

A proof of that claim is so straightforward that we leave it as an exercise to the reader. $\square$

Problem 7. Given two distinct points A and A' that are the inverses of each other with respect to a circle with a given radius r , locate the center of the corresponding inversion, O , with *negative* power.

Solution. One way to solve this problem is to investigate the algebraic side of the problem and then translate the found solution that is cast in the language of algebra into a solution that is cast in the language of geometry.

To that end, let the point O be the center of the inversion sought-after.

Then, we are given the distance $|AA'| = a$ between the points A and A' and we want to recover the distance $|OA| = x$ between, say, the point A and the sought-after center of inversion, O . According to **Condition 3** of **Definition 1** and our definition of inversion with respect to a circle we *negative* power, which implies that $|OA'| = (a - x)$, we have:

$$x \cdot (a - x) = r^2$$

from where it follows that:

$$x^2 - ax + r^2 = 0$$

Solving the above equation quadratic in x for x , we find that:

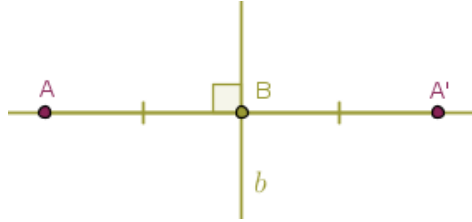
$$x_{1,2} = \frac{a}{2} \pm \sqrt{\left(\frac{a}{2}\right)^2 - r^2} \quad (2)$$

Analyzing the above relation, we see that all the algebraic material in it is constructible with Euclidean tools alone.

That is, a geometric equivalent of the square root in (2) is a side of a right triangle the lengths of whose hypotenuse, $a/2$, and the other side, r , are given because the length of the radius r is given and, because the length a is given, a geometric equivalent of $(a/2)$ is one half of a line segment of length a and any line segment can be bisected with Euclidean tools alone.

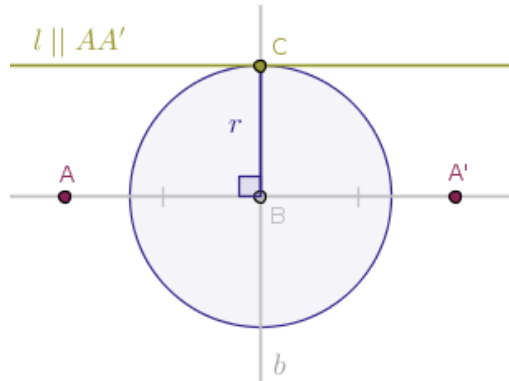
As such, translating the above algebraic or arithmetic sentence into a solution that is cast in the language of geometric objects, we have.

1. Draw a straight line through the given points A and A' , bisect the line segment AA' with a perpendicular bisector, b , in order to locate the point B on the line segment AA' such that $|AB| = |BA'| = a/2$ (Figure 4.1):

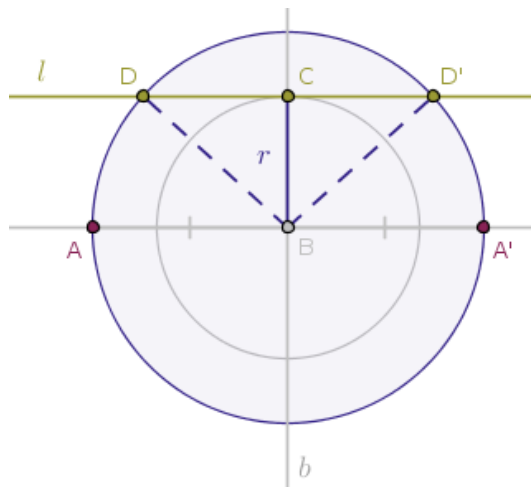


Here we reuse the entire step and the corresponding diagram from Step 1 of Problem 6, verbatim.

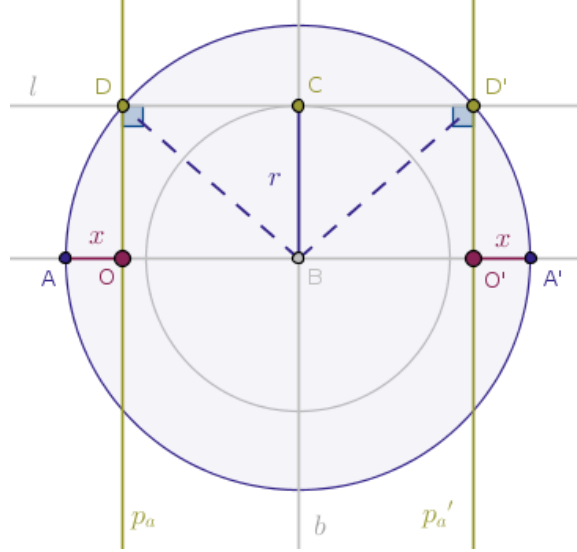
2. Draw a circle centered at the point B with the given radius r until that circle intersects the bisector b at the point C , draw a straight line, l , through the point C parallel to the straight line AA' (Figure 4.5):



3. Draw a circle centered at the point B with the radius $|BA| = |BA'|$ until that circle intersects the straight line l at the points D and D' (Figure 4.6):



4. Draw a straight line, p_a , perpendicular to l through the point D until p_a intersects the straight line AA' at the point O , draw another straight line, p'_a , perpendicular to l through the point D' until p'_a intersects the straight line AA' at the point O' (Figure 4.7):



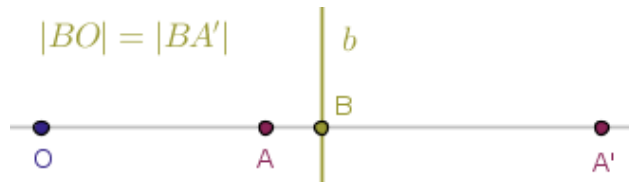
The points O and O' thus constructed are the sought-after centers of the two circles of radius r under the inversion with respect to which with *negative* power, the given points, A and A' , are the images of each other.

A proof of that claim is so straightforward that we leave it as an exercise to the reader. $\square$

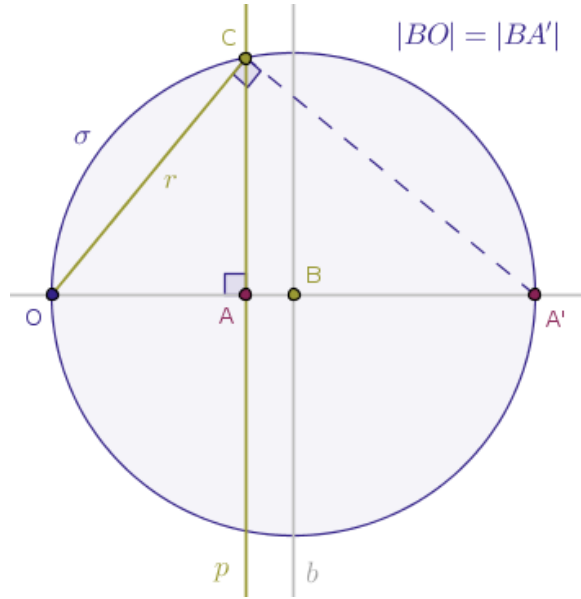
Problem 8. Given two distinct points A and A' that are the inverses of each other with respect to a circle with a given center O , construct the radius of inversion r with positive power.

Solution. While it is certainly possible to reuse the result of **Problem 6** to solve this problem, there is a simpler way that is based on the solution of **Problem 1**.

1. Draw a straight line through the collinear, by definition, points O , A and A' , bisect the line segment OA' with a straight line b until b intersects the straight line OA' at the point B such that $|BO| = |BA'|$ (Figure 4.8):



2. Draw a circle, σ , centered at the point B with the radius $|BO| = |BA'|$, draw a straight line, p , through the point A perpendicular to OA until p intersects the circle σ at the point C , connect the points O and C (Figure 4.9):



The length of the line segment OC thus constructed is the sought-after length of the radius r of the circle centered at the point O under the inversion with respect to which with positive power, the given points A and A' are the images of each other.

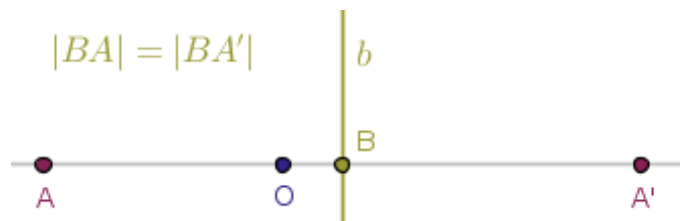
As a hint of a proof of that statement, that is left as an exercise to the reader, observe that because the angle OCA' subtends a diameter of the circle σ , that angle must be right. Which implies that the straight line CA' must be a tangent to the circle of inversion at hand. Recall the solutions of Problem 1 and Problem 3. $\square$

Think on your own whether it is possible to leverage the symmetric construction of the inverse point with positive power that we developed in Problem 5 in order to solve Problem 8.

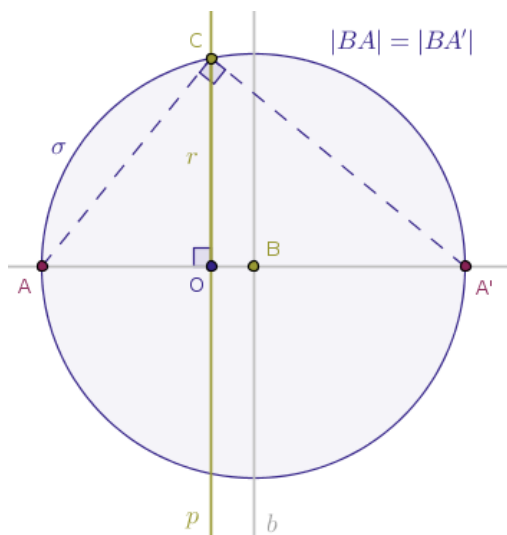
Problem 9. Given two distinct points A and A' that are the inverses of each other with respect to a circle with a given center O , construct the radius of inversion r with negative power.

Solution. While it is certainly possible to reuse the result of **Problem 7** to solve this problem, there is a simpler way that is based on the solution of **Problem 2**.

1. Draw a straight line through the collinear, by definition, points O , A and A' , bisect the line segment AA' with a straight line b until b intersects the straight line AA' at the point B such that $|BA| = |BA'|$ (Figure 4.10):



2. Draw a circle, σ , centered at the point B with the radius $|BA| = |BA'|$, draw a straight line, p , through the point O perpendicular to AA' until p intersects the circle σ at the point C (Figure 4.11):



The length of the line segment OC thus constructed is the sought-after length of the radius r of the circle centered at the point O under the inversion with respect to which with *negative* power, the given points A and A' are the images of each other.

As a hint of a proof of that statement, that is left as an exercise to the reader, observe that because the angle ACA' subtends a diameter of the circle σ , that angle must be right. The line segment CO is a height of the right triangle ACA' that emanates from the right-angle vertex C and falls on the triangle's base. Hence, Euclid's "*Elements*" Book 6 Proposition 8 is applicable to the right triangles AOC and COA' . $\square$

Theorem 6. The point, A' , that is inverse to a given point, A , under the inversion of a plane with respect to a circle $c(O, r)$ is unique.

Proof. A proof of this theorem follows directly from **Condition 3** of **Definition 1**. If we put $|OA'| = x$ and take the magnitudes of r^2 and $|OA|$ to be given and constant:

$$|OA| \times x = r^2$$

then the resultant equation for x will be an algebraic equation of degree one in a single variable over the field of real numbers:

$$|OA| \times x - r^2 = 0$$

and such equations always have a unique solution:

$$x = \frac{r^2}{|OA|}$$

provided, of course, that $|OA| \neq 0$, which geometrically means that the point A is distinct from the point O , which, as we agreed earlier, is the case.

The image of the center of inversion of a plane under an inversion with any power, positive or negative, is the point at infinity which is a planar equivalent of the Riemann Sphere's North Pole under the corresponding stereographic projection. $\square$

Note that this theorem is applicable to the inversion of a plane in a circle with both positive and negative powers.

Theorem 7. Under the inversion of a plane with respect to a circle $c(O, r)$ an even number of inversions of a point A with respect to c transforms A into itself.

Proof. It is sufficient to consider just two consecutive inversions of an original point A with respect to a circle c .

As such, assume that a point A is inverted into the point A' and the point A' is then inverted into the point A'' under the inversion of the same plane with respect to the same circle, c , with the same power.

Then, according to **Condition 3** of **Definition 1**, we have for both pairs of points:

$$|OA| \times |OA'| = r^2$$

$$|OA'| \times |OA''| = r^2$$

But the point A' is one and the same and, according to **Theorem 6**, it is unique.

Hence, for the distance of that point from the center of inversion, we have:

$$\frac{r^2}{|OA|} = \frac{r^2}{|OA''|}$$

Therefore:

$$|OA| = |OA''|$$

According to **Condition 1** of **Definition 1**, all the points O , A , A' and A'' must be collinear.

According to **Definition 2**, if the power of inversion is positive, the points A , A' and A'' must be located on the same side of the center of inversion, the point O .

According to **Definition 3**, if the power of inversion is negative, the center of inversion, the point O , must be located in between the points A and A' and in between the points A' and A'' .

Hence, from the above it follows that for either fixed power of inversion it must be the case $A'' = A$. $\square$

Note that the above theorem, which, by the way, is applicable to inversions with both positive and negative powers, can be stated slightly differently as follows:

if a point A' is the inverse of a point A under the inversion of a plane with respect to a circle c with either power, positive or negative, then the point A is the inverse of the point A' of the same plane with respect to c with the same power

We already made that conclusion at the end of Chapter 1, albeit ahead of the proof of Theorem 7.

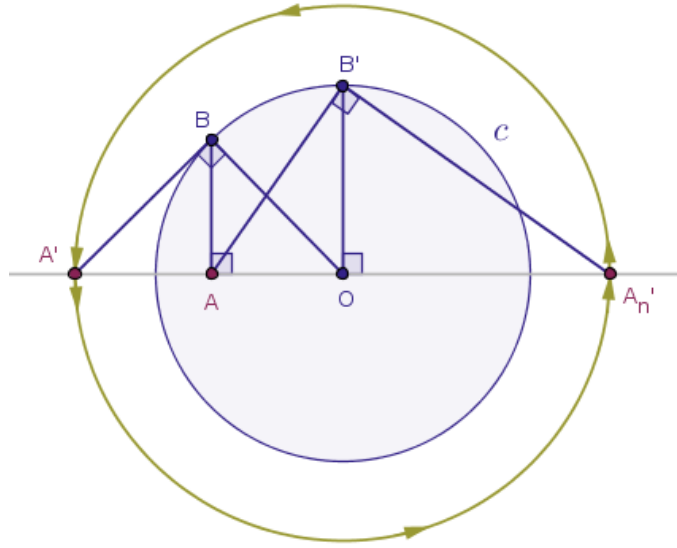
We conclude this chapter with an official restatement of the result that we knew to be true and correct from the solution of Problem 2.

Theorem 8. Inversion of a plane with respect to a circle $c(O, r)$ with negative power is equivalent to an inversion of the same plane with respect to the same circle c with positive power with a consequent 180-degree rotation of the parent plane about the center of inversion, O , and conversely.

Proof. Inverting a point A with respect to a circle of inversion, c , with positive power puts the image point A' of A on the same side of the center of inversion, O , while inverting a point A with respect to a circle of inversion, c , with negative power puts the image point A'_n on the opposite side of the center of inversion, O .

We have already dealt with this situation in Problem 2.

Essentially reusing the diagram from that problem, we have (Figure 4.12):



From the previous properties of inversion it follows, then, that regardless of where the original point A is located relative to the circle of inversion, c , **in/outside** of or **on** it, we have:

$$|OA| \times |OA'| = r^2 = |OA| \times |OA'_n|$$

Hence:

$$|OA'| = |OA'_n|$$

The points O and A are collinear with:

- the point A' , which is the inverse image of A with positive power and with
- the point A'_n , which is the inverse image of A with negative power

Hence, the points A' and A'_n must be collinear as well.

From where it follows that rotating the image point A' in the parent plane about the center of inversion, O , by half-a-turn will land the point A' directly on top of the image point A'_n and conversely, rotating the image point A'_n in the same plane about the center of inversion, O , by half-a-turn will land it directly on top of the image point A' . $\square$

Now that we understand how inversion of a plane with respect to a circle with positive and negative powers act on a single point, in the next chapter we will investigate how that inversion acts on straight lines.